

Truth tables

A truth table lists every input combination and the output F

Every row, one function

- Each row is one minterm index from all zeros through all ones
- AND is one only when every input is one
- OR is one when any input is one
- XOR is one when inputs differ
- Challenges in the lab ask you to match a described function
- Edit F in the table or the expression; both can stay in sync

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Truth-table builder

Set outputs or type an expression — the table and SOP/POS stay in sync. Expressions are evaluated by the public HDL engine (assign F = ...). Try the challenge set (AND through 2-bit equality), then save/print your work.

Starter example: two inputs, expression A & B fills the table. Challenges reset to a blank table when you Start.

Verilog: A & B

Load starter example

Challenges

PICK ONECore — Majority (3)

Set F = 1 when at least two of A, B, C are 1 (3-input majority).

StartShow hintNext challengeStop checkingIdle

Truth table

VARIABLES2 (4 rows)

NAMESA B

All 0All 1Clear X

A	B	F
0	0	
0	1	
1	0	
1	1	

Canonical forms

SOP (SUM OF PRODUCTS)AB

Use SOP as expression

POS (PRODUCT OF SUMS)

Truth table starter

Workbook practice

- In the workbook track, build a two-input AND and XOR table by hand
- Add a three-input majority row set
- Say how many rows you need for four variables
- Name one pitfall

Pitfalls to watch

- Do not skip rows or duplicate an index
- X in a cell means don't-care, not “pick any answer.” And remember
- Real verification still needs exhaustive or constrained cases beyond one hand table

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, write one full table and circle the minterms where F is one
- When you are ready, take the short quiz, then continue to Boolean laws